

Common patterns

Graphical linear algebra, the linear algebra course, and category theory —
and how to reason with tilings
(with machine-checked proofs in Lean 4)

Abstract

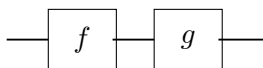
Three things that look different — the circuit diagrams of graphical linear algebra, the theorems of an undergraduate linear algebra course, and the arrows of category theory — are three dialects of one grammar. This note isolates the eight patterns they share, draws each of them, and points at the Lean theorem that proves it. It then does the same for a fourth dialect that is rarely put next to the others: *tilings and patterns*. Drawing a tiling is a legitimate way to reason about determinants, change of basis, Gaussian elimination, solution sets, rank and symmetry, and the last two sections collect the exam patterns and the big ideas that come out of it. Every picture in this note is backed by a theorem in the companion Lean 4 development in `RequestProject/`, which builds against Mathlib with no `sorry` and no added axioms. The two new modules are `RequestProject/CommonPatterns.lean` (the eight patterns) and `RequestProject/Tilings.lean` (the tiling material, up to the crystallographic restriction).

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1 One grammar, three dialects

A linear algebra course, a category theory course and the string-diagram calculus of graphical linear algebra all spend their time on the same short list of facts. They differ only in what they draw.

string diagram	commutative diagram	matrices
	$U \xrightarrow{f} V \xrightarrow{g} W$	$C = B A$

The dictionary is the following. Each row is proved, in the stated generality, in the file `CommonPatterns.lean`.

pattern	picture	textbook	Lean
composition	wires plug together	$(AB)C = A(BC),$ $IA = A$	<code>series_assoc,</code> <code>bare_wire_left</code>
functor	read a picture as a matrix	the matrix of a composite is the product	<code>toMatrix_series</code>
interchange	a picture has no reading order	$(A \otimes B)(C \otimes D) = AC \otimes BD$	<code>interchange_kronecker</code>
universal property	a box with a unique filler	extend a map linearly from a basis	<code>unique_extension_from_basis</code>
duality	bend a wire	$\langle Ax, y \rangle = \langle x, A^\top y \rangle$	<code>bend_wire, bend_series</code>
partition	cut a wire into strands	$A = \sum_{ij} a_{ij} E_{ij}$	<code>identity_is_tiled,</code> <code>matrix_is_tiled</code>
change of coordinates	redraw with new labels	similar matrices	<code>change_of_basis</code>
fibration	tile the domain in slices	solution = particular + kernel	<code>fibre_eq_kernel_translate</code>

2 The eight patterns, drawn

2.1 Pattern 1: composition (the category pattern)

Plugging wires end to end is associative, and a bare wire changes nothing. In a picture there are no brackets, which is exactly why nobody writes them.

$$\text{---} \boxed{f} \text{---} \boxed{g} \text{---} \boxed{h} \text{---} = \text{---} \boxed{f} \text{---} \boxed{g} \text{---} \boxed{h} \text{---}$$

Category theory: this is the definition of a category. *Textbook:* associativity of the matrix product, proved by exchanging two summation signs (`matrix_series_assoc`, and entry by entry in `SzaboMatrices.lean`). *Graphical linear algebra:* the fact that a diagram is a graph, not a term.

2.2 Pattern 2: functor (choosing a basis)

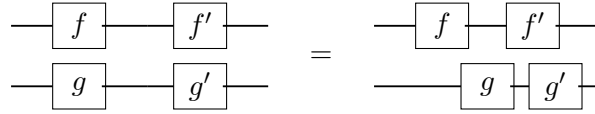
A basis turns a linear map into a matrix, and it does so *compatibly* with composition and identities. That compatibility is the single fact that makes matrix computations legitimate.

$$[g \circ f]_b^d = [g]_c^d [f]_b^c, \quad [\text{id}]_b^b = I.$$

Lean: `toMatrix_series, toMatrix_bare_wire`.

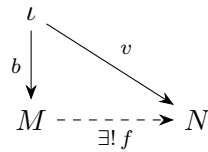
2.3 Pattern 3: interchange (a diagram has no reading order)

Two boxes drawn side by side may be composed first horizontally or first vertically; the results agree.



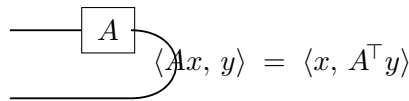
Category theory: $\otimes$ (or $\oplus$) is a bifunctor. *Textbook:* the mixed product rule $(A \otimes A')(B \otimes B') = AB \otimes A'B'$ for Kronecker products. *Lean:* `interchange_prod`, `interchange_kronecker`, `slide_past_wire`.

2.4 Pattern 4: universal property (say it on generators)



Every “define a linear map by prescribing its values on a basis” exercise is this triangle: the free module on the index set. Uniqueness is what lets a diagrammatic law be checked on generators only. *Lean:* `unique_extension_from_basis`, `eq_of_eq_on_basis`.

2.5 Pattern 5: duality (bending a wire)



Category theory: compact closedness — the cup and the cap and the two snake equations. *Textbook:* the transpose is the adjoint of the dot product, and $(AB)^T = B^T A^T$ because bending reverses the order. *Lean:* `bend_wire`, `bend_twice`, `bend_series`.

2.6 Pattern 6: partition (cut a wire into strands)

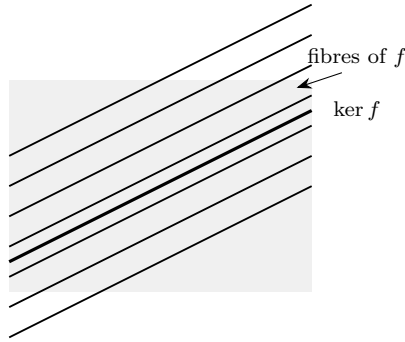
$$\sum_i E_{ii} = I, \quad A = \sum_{i,j} a_{ij} E_{ij}, \quad x = \sum_j x_j e_j.$$

This is already a tiling: the identity matrix is tiled by its diagonal cells and an arbitrary matrix by its rank-one cells. It is the computational content of “choose a basis”, and in category theory it is a biproduct decomposition $\text{id} = \sum_i \iota_i \pi_i$. *Lean:* `identity_is_tiled`, `matrix_is_tiled`, `vector_is_tiled`.

2.7 Pattern 7: change of coordinates

Relabelling the wires conjugates the box: $[f]_b^c = P[f]_{b'}^{c'}Q$ with P, Q the transition matrices (`change_of_basis`), and in the square case $B = X^{-1}AX$. Everything that survives conjugation — rank, trace, determinant, characteristic polynomial — is a property of the *picture* rather than of the coordinates; see `trace-loop.pdf` and `RequestProject/TraceLoop.lean` for the trace, where the invariance is the statement that a box may be slid around a closed loop.

2.8 Pattern 8: fibration (tile the domain)



The fibres $\{x : f(x) = c\}$ are the translates of $\ker f$: parallel tiles, one for each point of the image. Rank-nullity is the bookkeeping “(number of tiles) $\times$ (size of a tile) = size of the domain”:

$$\dim \operatorname{im} f + \dim \ker f = \dim V.$$

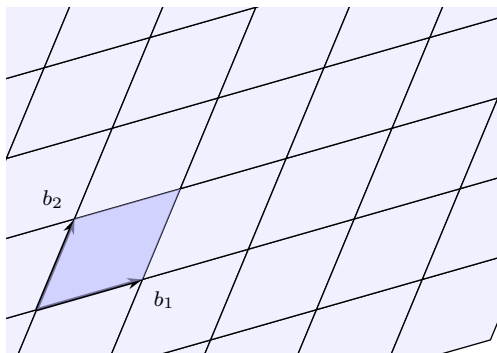
Lean: `fibre_eq_kernel_translate`, `fibres_disjoint_or_equal`, `fibres_cover`, `tiles_times_tile`, `solution_set_is_a_tile`.

3 Tilings and patterns as a reasoning device

Grünbaum and Shephard’s vocabulary translates into linear algebra almost word for word.

tiling	linear algebra	Lean
prototile	fundamental parallelogram of a basis	<code>ZSpan.fundamentalDomain</code>
the tiling (covers, no overlap)	every vector has unique coordinates mod the lattice	<code>plane_is_tiled_by_lattice</code>
area of the prototile	$ \det $	<code>tile_area_eq_abs_det</code>
re-tiling by a map	image of a set under a linear map	<code>area_scales_by_det</code>
shearing a tile	elementary row operation	<code>shear_preserves_area</code>
degenerate (flat) tile	singular matrix	<code>degenerate_no_tiling</code>
striped tiling by parallel lines	fibres of a linear map	<code>fibre_eq_kernel_translate</code>
symmetry group of the pattern	matrices preserving the lattice	<code>lattice_isometry_cases</code>
admissible rotation orders	1, 2, 3, 4, 6	<code>pow_twelve_eq_one</code>

3.1 A lattice tiles the plane; the tile has area $|\det|$



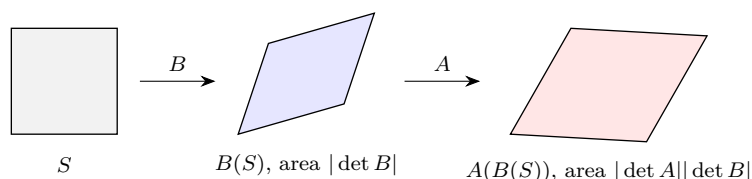
Formally: for a basis b of $\mathbb{R}^n$, every x lies in exactly one translate $v + D$, v in the lattice $\mathbb{Z}b_1 + \dots + \mathbb{Z}b_n$, where $D = \{\sum t_i b_i : 0 \leq t_i < 1\}$ (`plane_is_tiled_by_lattice`); all tiles are translates, hence of equal area (`tiles_are_congruent`); and

$$\text{area}(D) = |\det(b_1, \dots, b_n)| \quad (\text{tile_area_eq_abs_det}),$$

which in the plane is the familiar $|ad - bc|$ (`tile_area_fin_two`). *This is the definition of the determinant that makes every determinant exercise a picture.*

3.2 A linear map re-tiles space

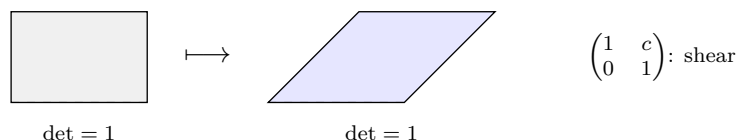
If f is linear then $\text{vol}(f(S)) = |\det f| \cdot \text{vol}(S)$ for *every* measurable S (`area_scales_by_det`). Two consequences that students are usually asked to prove algebraically become one-line pictures.



1. $\det(AB) = \det A \cdot \det B$: re-tile twice (`det_mul_is_retiling`).
2. $\det A = 0$ iff the tile is flat, iff some nonzero vector is killed, iff the columns are dependent (`degenerate_no_tiling`).

3.3 Gaussian elimination is a shear (Cavalieri's principle)

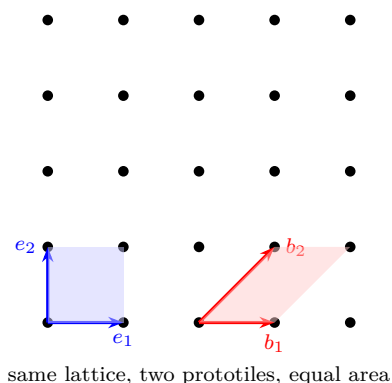
Adding a multiple of one row to another slides the tile parallel to itself: same base, same height, same area, so the determinant does not change.



Lean: `row_operation_preserves_det` (any size, any commutative ring) and `shear_det`, `shear_preserves_area` (the plane picture, with the measure). Swapping two rows reflects the tile ($\det \mapsto -\det$) and scaling a row stretches it ($\det \mapsto \lambda \det$): the three elementary operations are exactly the three ways of moving a tile.

3.4 Change of basis re-tiles the same plane

The *same* lattice can be tiled by different prototiles of the same area; the transition matrix is unimodular. This is the picture behind “similar matrices are the same map in different coordinates”.

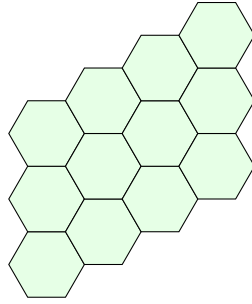


3.5 Symmetry: the crystallographic restriction

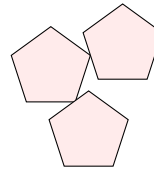
A symmetry of a lattice, written in a lattice basis, is an *integer* matrix. Its trace is therefore an integer. A rotation by θ has trace $2\cos\theta$, and $|2\cos\theta| \leq 2$; hence

$$2\cos\theta \in \{-2, -1, 0, 1, 2\}, \quad \text{so} \quad \theta \in \left\{ \pi, \frac{2\pi}{3}, \frac{\pi}{2}, \frac{\pi}{3}, 0 \right\}.$$

Only 1-, 2-, 3-, 4- and 6-fold rotations preserve a lattice: there is no fivefold wallpaper pattern. This is a two-line linear algebra proof of a theorem about tilings, and the whole of it is formalised (`two_cos_of_lattice_rotation`, `cos_of_lattice_rotation`).



6-fold: tiles



5-fold: gaps

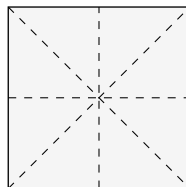
The algebraic version needs no geometry at all. From the 2×2 Cayley–Hamilton identity $A^2 = (\text{tr } A)A - (\det A)I$ one gets the recursion $s_{k+2} = t s_{k+1} - s_k$ for $s_k = \text{tr}(A^k)$, $t = \text{tr } A$; if $t \geq 3$ the s_k increase strictly, so $A^n \neq I$ for all $n \geq 1$ (`no_finite_order_of_trace_ge_three`). Applying this to A and to $-A$ gives $|\text{tr } A| \leq 2$ (`abs_trace_le_two`), and the five possible traces give five possible minimal polynomials, whence

$$A^{12} = I \quad (\text{pow_twelve_eq_one}).$$

Two corollaries that are fun to state: `no_fivefold_symmetry` (a unimodular integer matrix with $A^5 = I$ is the identity) and `hexagonal_rotation_pow_six` (the matrix $\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$ has order exactly 6, so the bound is attained — this is the 60° rotation of the triangular lattice written in the lattice basis).

3.6 The point group of the square lattice

Which integer matrices preserve lengths? $A^\top A = I$ forces $a^2 + c^2 = 1$, $b^2 + d^2 = 1$ and $ab + cd = 0$; over $\mathbb{Z}$ the first two equations leave four choices each and the third kills half of the sixteen combinations. Eight remain: the four rotations by multiples of 90° and the four reflections — the dihedral group of order 8.



4 rotations + 4 reflections = 8

Lean: `lattice_isometry_cases` and `isometry_set_eq` for the classification, `pointGroupSquare_card` for the count 8, and `pointGroupSquare_rotations_card` for the four rotations among them.

4 Exam patterns: what to draw, and why it is a proof

Exam pattern 4.1 (Compute a determinant). *Draw* the parallelogram (or parallelepiped) spanned by the columns. The determinant is its signed area. *Why it is a proof:* `tile_area_eq_abs_det`. Sanity check: two equal columns give a flat tile, so the determinant is 0.

Exam pattern 4.2 (Row reduce, keeping track of the determinant). *Draw* the tile before and after. Adding a multiple of a row shears it (area unchanged), swapping rows reflects it (sign flips), scaling a row stretches it. *Why:* `row_operation_preserves_det`, `shear_det`.

Exam pattern 4.3 ($\det(AB) = \det A \det B$). *Draw* the unit tile, its image under B , and then under A . Areas multiply. *Why:* `area_scales_by_det` plus `det_mul_is_retiling`.

Exam pattern 4.4 (When is a system solvable, and what is the solution set?). *Draw* the domain striped by the parallel translates of $\ker A$. If b is in the image, the solutions form exactly one stripe: one particular solution plus the kernel. *Why:* `solution_set_is_a_tile`, `fibre_eq_kernel_translate`.

Exam pattern 4.5 (Rank-nullity). *Draw* the same striped picture and count: the number of stripes is $\dim \text{im}$, the size of a stripe is $\dim \ker$. *Why:* `tiles_times_tile`.

Exam pattern 4.6 (Show that a quantity is basis-independent). *Draw* the diagram without labels; anything you can read off is invariant. For the trace, draw the closed loop and slide the boxes around it. *Why:* `change_of_basis`; and `Similar`, `trace_similar`, `det_similar`, `charpoly_similar` in `TraceLoop.lean`.

Exam pattern 4.7 (Prove an identity about tensor/Kronecker products). *Draw* the two boxes side by side and slide one past the other. *Why:* `interchange_kronecker`, `slide_past_wire`.

Exam pattern 4.8 (Define a map and check it is well defined). *Draw* the triangle: a map out of a free module is a function on the basis. *Why:* `unique_extension_from_basis`. For maps out of a quotient, the same triangle with the kernel condition: `quot_triangle` in `CategoryDiagrams.lean`.

Exam pattern 4.9 (Find all integer matrices of finite order / all lattice symmetries). *Draw* the lattice and ask which rotations map it to itself. *Why:* `abs_trace_le_two`, `pow_twelve_eq_one`, `lattice_isometry_cases`. Typical exam question: “show that a finite subgroup of $GL_2(\mathbb{Z})$ has order dividing 24” — the trace argument does it.

5 Big ideas

1. **Syntax is two-dimensional.** A formula is a line of symbols and must invent brackets; a diagram is a graph and gets associativity, units and the interchange law for free. What a string-diagram calculus buys is not new theorems but the elimination of bookkeeping.
2. **Every basic theorem is a normal-form statement.** “Every matrix is a product of elementary matrices”, “every diagram equals a matrix in reduced form”, “every morphism factors as epi followed by mono”: the same pattern, three languages.
3. **Invariance is the real content.** Rank, trace, determinant and characteristic polynomial are the quantities that survive relabelling; in the diagram they are the features you can see without reading the labels.
4. **Relations, not just maps.** Graphical linear algebra is *not* the calculus of matrices: it is the calculus of linear relations, which is why boxes may be run backwards and why it has the extra copy/discard/add/zero generators. Matrices are the sub-calculus of graphs of functions (`GraphicalLinearAlgebra.lean`, `DiagramDictionary.lean`).

5. **Counting is measuring is dimension.** The determinant measures a tile, the rank counts tiles in the image, the index of a lattice in another lattice is again a determinant. One idea, three names.
6. **Discreteness constrains symmetry.** Integrality of the matrix of a symmetry plus boundedness of the trace of a rotation gives the crystallographic restriction. A purely arithmetic fact ($2\cos\theta \in \mathbb{Z}$) rules out shapes.
7. **The same proof appears at three levels.** The proof that the trace is conjugation invariant, the proof that a closed wire may be slid, and the dinaturality of the categorical trace are literally the same argument in three notations.

6 Cool facts

- **No fivefold symmetry.** A unimodular integer 2×2 matrix with $A^5 = I$ is the identity (`no_fivefold_symmetry`); yet quasi-periodic Penrose tilings do have fivefold symmetry in a weaker, non-crystallographic sense — the theorem forbids exact translational periodicity, not the pattern.
- **Sixfold is attained.** $\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$ has order exactly 6 (`hexagonal_rotation_pow_six`, and its cube and square are not the identity).
- **Order twelve.** Every finite-order element of $SL_2(\mathbb{Z})$ satisfies $A^{12} = I$ (`pow_twelve_eq_one`).
- **The trace of a rotation is $2\cos\theta$,** so a rotation of a lattice has $2\cos\theta \in \{0, \pm 1, \pm 2\}$: the entire classification of wallpaper rotation orders is one integrality argument (`cos_of_lattice_rotation`).
- **Cayley–Hamilton in size two is a one-line computation** ($A^2 = (\text{tr } A)A - (\det A)I$, `cayley_hamilton_two`) and it powers the whole crystallographic argument through the trace recursion $s_{k+2} = ts_{k+1} - s_k$ (`trace_pow_rec`) — the same recursion that produces Chebyshev polynomials and Fibonacci-like growth.
- **The square lattice has exactly 8 symmetries, the “generic” lattice exactly 2** ($\pm I$): symmetry is the exception (`pointGroupSquare_card`).
- **A shear is invisible to area** but not to shape: Cavalieri’s principle and the invariance of the determinant under row operations are the same statement (`shear_preserves_area`).
- **Cutting a wire is a partition of unity.** $\sum_i E_{ii} = I$ is the smallest tiling in mathematics (`identity_is_tiled`).
- **Bending a wire twice does nothing** — the snake equation — and that is the entire reason $(A^\top)^\top = A$ and $(AB)^\top = B^\top A^\top$ (`bend_twice`, `bend_series`).

7 What is formalised, and where

file	contents
<code>CommonPatterns.lean</code>	the eight shared patterns: composition, functoriality of <code>toMatrix</code> , interchange (products and Kronecker), unique extension from a basis, bending a wire, partitions of the identity, change of basis, and the fibre tiling with rank-nullity
<code>Tilings.lean</code>	lattice tilings and fundamental domains, $\text{area} = \det $, the determinant as area scale factor, shears and row operations, the crystallographic restriction (algebraic and geometric form), no fivefold symmetry, the order-6 witness, and the point group of the square lattice
<code>GraphicalLinearAlgebra.lean</code> , <code>DiagramDictionary.lean</code>	diagrams as linear relations, and the dictionary to matrices and Penrose notation
<code>SzaboFields.lean</code> , <code>SzaboMatrices.lean</code> , <code>SzaboDeterminant.lean</code>	the textbook layer: number fields, matrices, the recursive determinant
<code>CategoryDiagrams.lean</code> , <code>ExistentialGraphs.lean</code>	commutative-diagram proofs; Peirce's alpha graphs
<code>ExamPatterns.lean</code> , <code>TraceLoop.lean</code>	standard exam patterns; the trace as a closed wire

Build with `lake build`; rebuild this note with `tectonic patterns.tex`.

References

- [1] B. Grünbaum and G. C. Shephard, *Tilings and Patterns*, Freeman, 1987.
- [2] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer.
- [3] L. Szabó, *Bevezetés a lineáris algebrába*, Polygon, Szeged, 2006.
- [4] P. Sobociski, *Graphical Linear Algebra*, <https://graphicallinearalgebra.net>.